

AWAH KENRIQUE ANYERE NGWA

Java Software Engineer

Douala, Cameroon • +237 672354082 • ngwakenri2016@gmail.com

[LinkedIn](#) • [GitHub](#) • [Portfolio](#)

PROFESSIONAL SUMMARY

Java Software Engineer with 5+ years delivering enterprise backend systems on Java 21 and Spring Boot 3 for government, financial and agricultural clients. Owns delivery end to end, from authoring Software Requirements Specifications and low-level architecture designs through implementation, testing and production support. Writes unit, feature and integration tests under Test-Driven Development, containerises services with Docker, and builds Continuous Delivery pipelines that cut deployment time by 30%. Tunes and profiles against non-functional targets, achieving 40% faster queries and ~50% lower API latency. Delivers within Scrum sprint rituals and traditional SDLC, leads code reviews, and mentors developers on clean architecture and open standards.

TECHNICAL SKILLS

Languages	Java 21 (LTS), Java 17, SQL, JPQL
Frameworks	Spring Boot 3, Spring MVC, Spring Data JPA, Spring Security, Spring Cache, Spring Retry, Spring Boot Actuator, Hibernate, Jakarta Validation
Architecture	Microservices, Modular Monolith, Domain-Driven Design, Bounded Contexts, RESTful API Design, Event-Driven Architecture, Apache Kafka, Resilience4j
Design and Docs	Software Requirements Specification (SRS) Authoring, Low-Level Design, Architecture Decision Records, C4 and UML Modelling, Data Modelling, OpenAPI 3.1, Swagger, Requirements Traceability
Testing	Test-Driven Development, JUnit 5, Mockito, Spring Boot Test, Integration and Feature Tests, Test Stubs, Test Automation, JaCoCo Coverage, Selenium, Katalon, Postman
DevOps	Docker, Docker Compose, Kubernetes, CI/CD, Jenkins, GitHub Actions, Git, Maven, Gradle, Liquibase, Nginx, Linux, AWS (EC2, S3)
Performance	Application Profiling, Performance Tuning, Query Optimisation, Connection Pooling, Caching Strategy, Prometheus, Micrometer, Grafana, ELK Stack
Data and Security	PostgreSQL 16, MySQL, Redis 7, Database Indexing, JWT, OAuth 2.1, OpenID Connect, Keycloak, Role-Based Access Control, TLS 1.3
Mobile and AI	React Native, Offline-First Sync, Push Notifications, Predictive Analytics Integration, Machine Learning Model Serving via REST
Delivery	Agile and Scrum, Sprint Planning, Backlog Refinement, Retrospectives, Traditional SDLC, Code Review, Mentoring, Stakeholder Workshops

PROFESSIONAL EXPERIENCE

Freelance Software Engineer

January 2024 – Present

Independent Consultant • Douala, Cameroon

- Authored the full Software Requirements Specification and system architecture for SHEMOLDS, a Cameroon Ministry of Health digital health platform, covering functional, non-functional, security and integration requirements with end-to-end traceability.
- Built the SHEMOLDS backend as a modular monolith on Spring Boot 3 and Java 21, applying Domain-Driven Design across 11 bounded contexts.
- Designed versioned REST APIs documented with OpenAPI 3.1, secured by JWT and OAuth 2.1 through Keycloak and Spring Security with role-based access control.
- Developed the React Native student mobile application with offline-first SQLite storage and version-based conflict resolution for intermittent connectivity.
- Integrated an AI health status prediction service that scores member health risk from claims and check-up history to flag preventive care needs.
- Implemented claims adjudication enforcing a 5-working-day SLA, with Hibernate Envers audit trails and Liquibase schema migrations.
- Integrated MTN Mobile Money and Orange Money payment APIs using Spring Retry backoff and Resilience4j circuit breakers.
- Cut database load with a Redis 7 cache-aside strategy and tiered TTLs, making dashboard KPIs load instantly.
- Containerised the stack with Docker and Docker Compose behind Nginx, instrumented with Prometheus, Grafana and ELK logging.
- Delivered the CNERSH research ethics platform for the Cameroon Ministry of Health, cutting administrative processing time by 60%.
- Built the Kombe Farmers Wallet aquaculture platform with AI weather automation and fish growth prediction, serving 500+ agricultural workers.

Quality Assurance Engineer

June 2024 – March 2025

Climate Wavers • Nairobi, Kenya (Remote)

- Automated end-to-end testing for a nationwide disaster management system, increasing test coverage by 85%.
- Resolved 50+ critical production defects, shortening sprint turnaround by 20% and enabling a zero-downtime deployment.
- Authored regression and integration test suites using TDD, standardising QA protocols across the engineering team.

Software Engineer

September 2020 – May 2023

Nasia Technology PLC • Douala, Cameroon

- Developed enterprise financial software in Java Spring Boot, scaling to thousands of daily active users.
- Refactored legacy modules and optimised SQL queries, improving processing speed by 40% through profiling and debugging.
- Automated bi-weekly release pipelines with CI/CD, cutting deployment time by 30%.
- Led code reviews, wrote low-level designs and technical specifications, and mentored junior developers.

Computer Science & STEM Educator

2024 – 2026

Cambridge High School & Divine Favour High School • Douala, Cameroon

- Mentored 60+ students in advanced Computer Science, algorithms and data structures.

SELECTED TECHNICAL PROJECTS

SHEMOLDS Digital Health Management Platform | Java 21, Spring Boot 3, React Native, PostgreSQL, Redis, Docker

Douala, Cameroon

- Cameroon Ministry of Health platform delivering pooled health coverage, a savings scheme and preventive health education to students nationwide.
- Authored the complete SRS and modular monolith architecture, decomposing the domain into 11 bounded contexts with Domain-Driven Design.
- Developed the React Native mobile application with offline-first sync, mobile money payments, claims submission and push notifications.
- Integrated AI health status prediction scoring member risk from claims and check-up history to trigger preventive interventions.
- Automates the member lifecycle end to end: enrolment, premium collection, claims adjudication, investment payouts and impact reporting.

CNERSH Research Ethics Platform | Java, Spring Boot, PostgreSQL, REST APIs

Yaoundé, Cameroon

- Cameroon Ministry of Health platform digitising submission and ethical review of human health research protocols.
- Multi-stage reviewer workflow with role-based access control and full audit logging on a Spring Boot REST backend.
- Replaced paper-based institutional review, cutting administrative processing time by 60%.

Kombe Farmers Wallet Web Application | Java, Spring Boot, PostgreSQL, Redis, IoT Sensors

Douala, Cameroon

- Aquaculture and agricultural management platform handling real-time financial transactions for 500+ regional farmers.
- Built AI weather automation forecasting local conditions to schedule feeding, harvesting and pond management activities.
- Developed a fish growth prediction model correlating water quality analysis, feeding cycles and temperature to project yield and optimise feed cost.
- Cut backend API response latency by ~50% with Redis caching, improving resource tracking accuracy by 100%.

Climate Wavers Disaster Management Application | Selenium, Katalon, TDD, CI/CD

Nairobi, Kenya

- Nationwide application for reporting climate disaster events and coordinating emergency response.
- Built the automated test framework across citizen reporting, alerting and administrative workflows, raising coverage to 85%.
- Integrated the suite into the delivery pipeline, catching 50+ defects before release.

Automated School Report Engine | Java, Spring Boot, Desktop Application

Douala, Cameroon

- Desktop application automating student grading and end-of-term report generation.
- Eliminated manual clerical errors and cut administrative data entry time by 80%.

EDUCATION

B.Sc. Computer Science & Software Engineering — 3.10 GPA

2022 – 2023

Higher Institute of Technology, Business and Management Sciences

HND Computer Science & Software Engineering — 2.87 GPA

2020 – 2022

Higher Institute of Technology, Business and Management Sciences

CERTIFICATIONS

- Professional Software Engineer — HackerRank, 2025
- Info Security & Quality Assurance — freeCodeCamp, 2023

- REST API Intermediate — HackerRank, 2024
- QA Certificate of Excellence — Climate Wavers, 2024
- Backend Development & APIs — freeCodeCamp, 2023
- Full Stack Development — freeCodeCamp, 2023

REFERENCES

Mr. Nji Edwin Ngwa — Senior Software Engineer, Accenture, Virginia, USA
+1 (540) 841-1552 • edwin.n.ngwa@gmail.com